

JOSHUA CROSS

GIS Analyst | B.A. Geography | Army Veteran | Federal Project Manager
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PROFESSIONAL SUMMARY

GIS analyst with a B.A. in Geography and GIS-focused coursework (summa cum laude), plus over four years of GIS-related technical and IT experience spanning geodatabase management, SQL-backed data systems, spatial data QA/QC, and infrastructure mapping, alongside a decade of federal project and field operations experience. Builds and maintains geodatabases, produces cartographic and web map products, and performs terrain, raster, and volumetric analysis on LiDAR and SfM-derived surfaces. Experienced with GPS and mobile field data collection, QA/QC of technical data prior to federal submission, and automation of data workflows through scripting. Works independently across multiple concurrent projects and communicates technical findings clearly to non-specialist stakeholders. Former TS/SCI and DOE Q clearance holder. Spokane-based, field-hardened, and available for travel.

CORE COMPETENCIES

- **Spatial Data Management:** Esri geodatabase creation and editing, attribute schema design, metadata, coordinate systems, projections and datum transformations
- **Data Quality Control:** structured QA/QC review of technical data and documentation, error identification and correction prior to submission, reconciliation of field data against records
- **Mapping & Visualization:** ArcGIS Pro and QGIS map production, thematic and choropleth mapping, poster-scale cartography, dashboards and StoryMaps
- **Spatial Analysis:** terrain and raster analysis, geomorphic change detection, volumetric estimation from elevation surfaces, DEM analysis and SfM point cloud processing
- **Field Data Collection:** ArcGIS Field Maps, Survey123, GPS-based survey and site data capture, hydrological field measurement
- **Remote Sensing:** Sentinel and Landsat imagery analysis in Google Earth Engine
- **Scripting & Automation:** JavaScript/Node.js development, R (terra, sf, rgl), SQL querying, working familiarity with Python; REST API integration and automated data pipelines
- **Data & Reporting Tools:** Excel formulas, lookups, and tabular data cleanup; technical documentation, stakeholder presentations, and end-user training

EDUCATION

Penn State - Graduate Certificate in Geospatial Intelligence Analytics (In Progress, Expected Summer 2027)

San Diego State University - B.A. in Geography, 2026. Summa cum laude, GPA 3.87. Coursework concentrated in GIS

Norco College - A.S. in Industrial Automation, 2024

APPLIED GIS PROJECTS

- **SIGACTS Conflict Data Visualization:** normalized 1.8M+ significant-activity records from Iraq and Afghanistan into a spatial database and built an interactive mapping platform for trend and pattern analysis.
- **Aquaplex Canyon Terrain Analysis:** combined a USGS LiDAR-derived DEM with drone Structure-from-Motion surfaces to model erosion and produce a 191,139 m³ volumetric estimate of the eroded valley, published as an ArcGIS StoryMap with an interactive 3D model.
- **Alvarado Creek Discharge Survey:** conducted GPS-located field measurement of stream discharge, built the supporting geodatabase, and ran QA/QC on collected field data prior to analysis.
- **California Job Mapping Application:** full-stack web map (Node.js, Express, MongoDB, Leaflet) integrating a live external API, with automated data ingestion and cloud deployment.
- **Precinct-Level Election Choropleth:** joined tabular results to precinct geometry, resolved boundary and attribute mismatches, and produced publication-quality thematic cartography.

PROFESSIONAL EXPERIENCE

Project Manager, Venergy Group LLC

Oct 2017 - Mar 2022 | Federal Projects Nationwide

- Managed 25+ federal contracts covering utility metering and solar system maintenance, independently coordinating site data, schedules, and deliverables across facilities nationwide
- Served as project manager for the Department of Veterans Affairs nationwide energy monitoring system; Schneider Electric certified to configure and deploy the platform

- Queried and reported on location-referenced metering data (by site, building, and address) from a SQL Server-backed monitoring platform (Schneider Electric PME), standardizing data collection and performance tracking across facilities nationwide
- Assessed, repaired, and managed an 11.4MW PV portfolio for California VA hospitals, maintaining a geospatial inventory of asset locations and performance data across the portfolio
- Led QA/QC reviews of technical specifications and site documentation, maintaining compliance with federal environmental and reporting standards
- Produced and maintained infrastructure schematics, site distribution maps, and geospatial asset location records to support portfolio management and field operations reporting

Senior WMD/CBRNE Instructor, *Chimera Enterprises International* Jan 2016 - Oct 2017 | Global Deployment

- Integrated spatial awareness and facility mapping into site-specific threat assessments delivered at U.S. embassies worldwide; instructed 8,000+ personnel.
- Produced mapped countermeasure overlays for embassy security planning, translating physical site surveys into structured, spatially-referenced documentation.

EOD Team Leader, U.S. Army

Aug 2011 - Aug 2015 | U.S., South Korea, Vietnam

- Directed POW/MIA recovery fieldwork in Vietnam using site grid analysis, terrain interpretation, and systematic field documentation
- Managed accountability for \$3.5M in specialized equipment across multi-site operations; responded to 15+ UXO incidents under field conditions

Infantryman, U.S. Army

Sep 2005 - Aug 2011 | Schofield Barracks, HI / Operation Iraqi Freedom

- Led combat patrols during Operation Iraqi Freedom, conducting 300+ missions and applying terrain analysis, route planning, and real-time situational mapping to mission execution

CERTIFICATIONS

Schneider Electric Project Deployment for System Engineers • Nalco Water Environmental Water Sampling & Handling • Basic Instructional Design • EOD Basic & Team Leader Certifications • SDSU AI & Machine Learning Micro-Credential (2025) • Valid driver's license